

fkidx (runner patch) – guide

Revision: 1 Last modified: 2026-09-26T13:40:00Z

Source: `constitution/scripts/codegraph/runner_patches/fkidx1.py` (107 lines). It is a **module driven by** `fk_index_patch.py`, not a command: nothing in it is run directly. Citations are `fkidx1.py:<line>` unless another file is named. Machinery guide: [fk_index_patch](#).

Overview

Stock CodeGraph 1.6.0 speeds up bulk indexing by **dropping secondary indexes** before a bulk window (three lists in `lib/dist/db/index.js` :

`BULK_PARSE_INDEX_NAMES` , `BULK_REF_INDEX_NAMES` , `BULK_EDGE_INDEX_NAMES`) and recreating them afterwards. It also runs `INSERT OR REPLACE INTO nodes with foreign_keys=ON` . A `REPLACE` of an existing node id deletes the old row, which fires `ON DELETE CASCADE` on every child table with a foreign key to `nodes(id)` ; if the child's key column has no index, each cascade is a **full table scan** (module docstring, lines 1-8).

`fkidx1` edits `lib/dist/db/index.js` in the private runner copy so that the FK child-key indexes are **removed from the drop lists** (they stay in place through the bulk windows). It never edits the shared install (that is the machinery's job, see the parent guide).

The keep-set is not hard-coded: it is **derived from the target's own** `schema.sql` (lines 26-37, 51-71), so it adapts to upstream schema changes and reports `NOT_NEEDED` when the hazard is not there.

Prerequisites

- Files read from the package (lines 16-18): `lib/dist/db/index.js` , `lib/dist/db/schema.sql` , `lib/dist/db/queries.js` .
- Python 3 standard library only (`re`), plus `runner_patches/common.py` (`NotNeeded` , `Refuse` , `PatchPlan`).
- Loaded by `runner_patches.load()` ; registered first (`runner_patches/__init__.py:17`), default set member (not opt-in).

Usage

Only through `fk_index_patch.py` :

```
python3 scripts/codegraph/fk_index_patch.py --patches fkidx1 --dst <scratch dir> [--src <package dir>]
```

`NOT RUN` in this form (it copies the ~282 MB package; see the parent guide for the tiny-fixture build that was run). What **was** run (read-only planning, no copy, no write):

- `VERIFIED` (ran: `python3` with `sys.path` at `scripts/codegraph`; `FileView` over a scratch dir holding stock 1.6.0 `db/index.js` + `schema.sql` + `queries.js`; `fkidx1.plan(view)`; saw: `kept_indexes = [idx_edges_target_kind, idx_unresolved_from_node]`; list sizes `BULK_PARSE 15 -> 14`, `BULK_REF 5 -> 4`, `BULK_EDGE 4 -> 3`) . So on stock 1.6.0 the patch keeps exactly two indexes and touches all three lists (a kept index can appear in more than one list).
- The planned `db/index.js` passes `node --check` with the package's bundled node (part of the parent guide's verification, with a broken-file control that exits 1).

Behaviour (source-traced)

`plan(view)` (lines 97-107):

1. `compute_patch` (lines 51-71):

- NOT_NEEDED if `queries.js` no longer contains `INSERT OR REPLACE INTO nodes` (line 52-53);
- NOT_NEEDED if `db/index.js` no longer contains `foreign_keys = ON` (regex `foreign_keys\s*=\s*ON`, lines 54-55);
- `derive_keep_set` (lines 26-37): walks `CREATE TABLE IF NOT EXISTS ...`; blocks in `schema.sql`, collects `FOREIGN KEY (col) REFERENCES nodes(id) columns`, and every `CREATE [UNIQUE] INDEX IF NOT EXISTS` with its leading column; NOT_NEEDED if there is no FK to `nodes(id)` at all (line 31-32);
- `parse_lists` (lines 40-48): parses the three `static BULK_*_NAMES = [...]` lists; **Refuse** if any of the three is missing or empty ("upstream shape changed");
- for every FK child column, if some index led by that column is **never dropped** in any list, nothing to do (lines 60-62); else it must have an index candidate in `schema.sql` (else **Refuse**, line 64-65) and the candidate with the fewest columns (ties by name) is added to the keep set (lines 66-67);
- NOT_NEEDED if the keep set ends up empty (lines 69-70).

2. `patch_text` (lines 74-95): for each kept index, removes its quoted name from every list that holds it; **Refuse** if the name is in no list (line 79-80), does not appear exactly once in a list (lines 86-88), or if a list would become empty (lines 92-94).

3. The plan records the edited file, one anchor per touched list, and `detail = {kept_indexes, lists_before, lists_after}` (lines 100-106);
`fk_index_patch.py` copies those into the legacy top-level receipt fields.

Outputs: an edited text for `lib/dist/db/index.js` only. Side effects: none in this module (pure function of the three files' text).

Edge cases

All VERIFIED (ran: planning against copies of the stock 1.6.0 files in a scratch directory) :

Condition	Result
Stock 1.6.0	APPLIED, keep = idx_edges_target_kind , idx_unresolved_from_node
Plan again on the already-patched text	NOT_NEEDED: every FK child column is already served by a never-dropped index (not an error, unlike the other patches)
INSERT OR REPLACE INTO nodes replaced by INSERT INTO nodes in queries.js	NOT_NEEDED: queries.js no longer uses INSERT OR REPLACE INTO nodes
foreign_keys = ON changed to OFF in db/index.js	NOT_NEEDED: db/index.js no longer enables foreign_keys
REFERENCES nodes renamed in schema.sql	NOT_NEEDED: schema.sql has no FOREIGN KEY ... REFERENCES nodes(id)
BULK_EDGE_INDEX_NAMES renamed in db/index.js	Refuse: db/index.js: BULK_EDGE_INDEX_NAMES not found/empty – upstream shape changed

Known limitation (from tests/test_patch_machinery.py , run 2026-09-26): the list parser does not assert that each static BULK_*_NAMES declaration is unique, so a commented-out duplicate or a duplicated declaration is not refused (cases P1b/P1c FAIL today; this is the tool's own RED test). See the parent guide, "Known discrepancies".

Internal behaviour / oracle

The external oracle for this patch is `fk_cascade_probe.py` (module header lines 6-7): it demonstrates the cascade full-scan hazard on a stock database and that the patched runner removes it. It is a separate tool with its own guide ([fk_cascade_probe](#)); `codegraph_safe.sh` runs it to prove stock safe when the machinery exits 3 (NOT_NEEDED for every patch), per the parent guide.

Tests

`fkidx1` has no dedicated test file. It is exercised by `tests/test_patch_machinery.py` (`fkidx1` build cases, P1-P4) and referenced by `tests/test_bulk_classify_mutate.py`, `tests/test_mcp_readonly_mutate.py`, `tests/test_unit_safe.sh` (those name the id in registry/selection arguments). `owed_report.py` counts it as tested for that reason. FINDING: there is no behavioural test of the *derivation* logic (keep-set for other schemas) in the tree; the checks in the Edge cases table above were run ad hoc for this guide.

Related scripts

[fk_index_patch](#) (the machinery), [fk_cascade_probe](#) (oracle), [codegraph_safe](#) (consumer; default patch set `fkidx1, resolve1`), siblings [lockfix1](#), [datafrag1](#), [resolve1](#), [resolve2](#), [mcpro1](#).

Last verified

2026-09-26, against stock CodeGraph 1.6.0 files copied to a scratch directory; no writes to the real runner cache or to any `.codegraph` index.